Tarun P. M.

AI/ML Engineer | Deep Learning | Computer Vision | Speech | Agentic Systems
Germany | ta***@gmail.com | LinkedIn | GitHub

PROFILE

AI/ML engineer with 1.5+ years of research experience in event-based computer vision, speech emotion recognition and deep learning, complemented by hands-on development of LLM-powered agents and workflow automation. Experienced in audio feature extraction, CNN and BiLSTM architectures, model evaluation, ablation studies, GPU/HPC training, Google Cloud deployment and human-in-the-loop agent design. Currently developing a self-hosted AI job discovery platform with local LLM integration, structured evaluation, validation and automated workflows.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: PyTorch, TensorFlow, Scikit-learn
Cloud and Infrastructure: Git, Vertex AI, GitLab CI
Speech and Computer Vision: MFCC

EXPERIENCE

Master's Thesis Research — Event-Based Lip Reading with Mamba Jan 2025 – Aug 2025
Forschungszentrum Jülich Aachen, Germany

- Developed a Mamba-based deep learning pipeline for event-driven lip reading, integrating selective state-space modeling with the DVS-Lip dataset to evaluate temporal dependencies in neuromorphic vision.
- Implemented voxelization and preprocessing pipelines for asynchronous camera events, converting sparse spatiotemporal data into structured representations suitable for neural network training.
- Built a custom hybrid sequence-generation framework combining event-based lip-reading data with artificial grammar sequences to evaluate long-range dependencies and controlled sequence complexity.
- Conducted experiments across optimizer configurations, model depths, and grammar ambiguity levels, achieving 63.7% accuracy on DVS-Lip and 78–80% test accuracy on a separate hybrid sequence-prediction task.

Technologies: Python, PyTorch, Mamba, State-Space Models, CUDA, GPU/HPC, NumPy, Event-Based Vision, GitLab

Research Assistant — Speech Emotion Recognition Jun 2024 – Dec 2024
Universität Siegen Siegen, Germany

- Developed a speech emotion recognition pipeline using Python and Librosa, preprocessing 1,440 RAVDESS audio recordings and extracting MFCC features for emotion classification.
- Implemented and evaluated SVM, Random Forest, CNN, and CNN-BiLSTM models, comparing traditional machine learning approaches with deep learning architectures.
- Achieved 96.41% test accuracy and 96.41% F1-score with an augmented CNN-BiLSTM model, evaluating performance using precision, recall, and confusion matrices.

Technologies: Python, TensorFlow, Keras, Scikit-learn, Librosa, MFCC, CNN, BiLSTM, NumPy, Pandas

PROJECTS

AI Job Hunt Platform

In active development | 2026

- Integrated Ollama with Qwen3 8B to evaluate job descriptions against a structured candidate profile, generating job-fit scores, skill-gap assessments and structured recommendations.
- Implemented an LLM evaluation pipeline with JSON parsing, deterministic score classification, decision-mismatch tracking, structural validation and error handling for inconsistent model outputs.
- Built sequential job-processing workflows with conditional routing and validation gates, preventing structurally invalid evaluations from entering downstream processing.
- Integrated n8n Data Tables for persistent job records and evaluation results; developing automated resume tailoring, PDF generation and human-in-the-loop application review.

Technologies: n8n, JavaScript, Ollama, Qwen3 8B, Docker Compose, Gotenberg, REST APIs, n8n Data Tables, LLM Evaluation, AI Agents, Git

Ambient Expense Agent

Human-in-the-Loop Agent on Google Cloud

Completed

- Deployed a Gemini agent using Google ADK 2.0 that auto-approves low-value claims and escalates high-value claims to human review through Vertex AI Agent Runtime and Pub/Sub.
- Built a FastAPI dashboard on Cloud Run with session persistence, resumable execution, IAM and dead-letter handling; adversarial testing confirmed escalation during a simulated prompt-injection attempt.

Technologies: Google ADK 2.0, Gemini, Vertex AI Agent Runtime, Pub/Sub, FastAPI, Cloud Run, IAM

EDUCATION

Master of Science in Mechatronics — Oct 2025

Universität Siegen, Germany — Grade: Good

Bachelor of Technology in Mechanical Engineering — Apr 2019

Vardhaman College of Engineering, India

LANGUAGES

English — business fluent | German — B1, actively improving | Hindi — fluent | Telugu — native